

# Tutorial 7 – Case Study: Data-Driven Decisions at Amazon

## (Take-home tutorial)

### Objective

To understand how Amazon integrates big data and business intelligence tools into its decision-making processes, and to apply concepts from the lecture, such as decision types, BI functions, and decision-support systems, to a real-world organizational context.

### Duration

1 – 1.5 hours

### Activity Instructions

1. Read Article (20 minutes)

Students should read the article:

[The Power of Data: How Amazon Utilizes Big Data to Drive Sales](#) — This article provides a foundation for understanding how Amazon applies data-driven systems to support a wide range of decisions, from daily operational tasks to high-level strategic planning.

[Amazon's Business Intelligence Tools: Advanced Techniques for Analyzing and Visualizing Data](#) — This article highlights the BI tools Amazon employs and explores how different management levels interact with these systems.

Understand how Amazon's approach reflects the decision-making concepts introduced in the lecture.

2. Answer the following questions:

- a. At which decision points does Amazon apply its data-driven systems?
- b. Which decisions at Amazon are structured, semi-structured, or unstructured? ( L9 pp.8-9)
- c. What types of Business Intelligence (BI) functions does Amazon use? ( L9 pp.18-21)
- d. Which decision-support systems does Amazon use?
  - Decision-Support Systems (DSS) (L9 pp.27)
  - Executive Support Systems (ESS) ( L9 pp.28-30)

3. Individual Submission (On Canvas as per the deadline indicated for Tutorial 7)

1. Each student must individually submit answers to the questions in a **PDF file of no more than two pages**, named:  
T7\_<full-name>.pdf(e.g., T7\_xiaoming.pdf).
2. Students must submit their work **by the deadline to receive score for this tutorial.**
3. **Delayed submissions will NOT get any score.**

## **Quick Reading Aid:**

### **Background**

Amazon, one of the world's largest e-commerce and cloud computing companies, has built its success on the strategic use of data. As its operations scale globally, Amazon collects and processes massive volumes of data in real time—from customer behavior and transaction records to inventory levels and delivery logistics. This data serves as the backbone for a wide range of decision-making processes at every level of the organization. To harness this information effectively, Amazon has developed sophisticated business intelligence (BI) and analytics systems, enabling fast, automated, and increasingly precise decision-making across business functions.

### **Overview**

The first article explores how Amazon leverages big data to enhance its e-commerce operations. It describes how customer behavior, transaction data, and historical trends are used to optimize key business processes, including personalized product recommendations, dynamic pricing, and customer segmentation. The article also illustrates how data insights are integrated into the user experience to improve engagement and increase sales conversion.

The second article provides an inside look at Amazon's use of business intelligence tools and platforms. It introduces the infrastructure, and systems Amazon employs to collect, store, analyze, and visualize large-scale data. The article highlights how these tools are used by different business units to monitor performance, detect trends, and support data-driven decision making across various organizational levels.